

Continuous Quantum Error Correction with ML Decoders

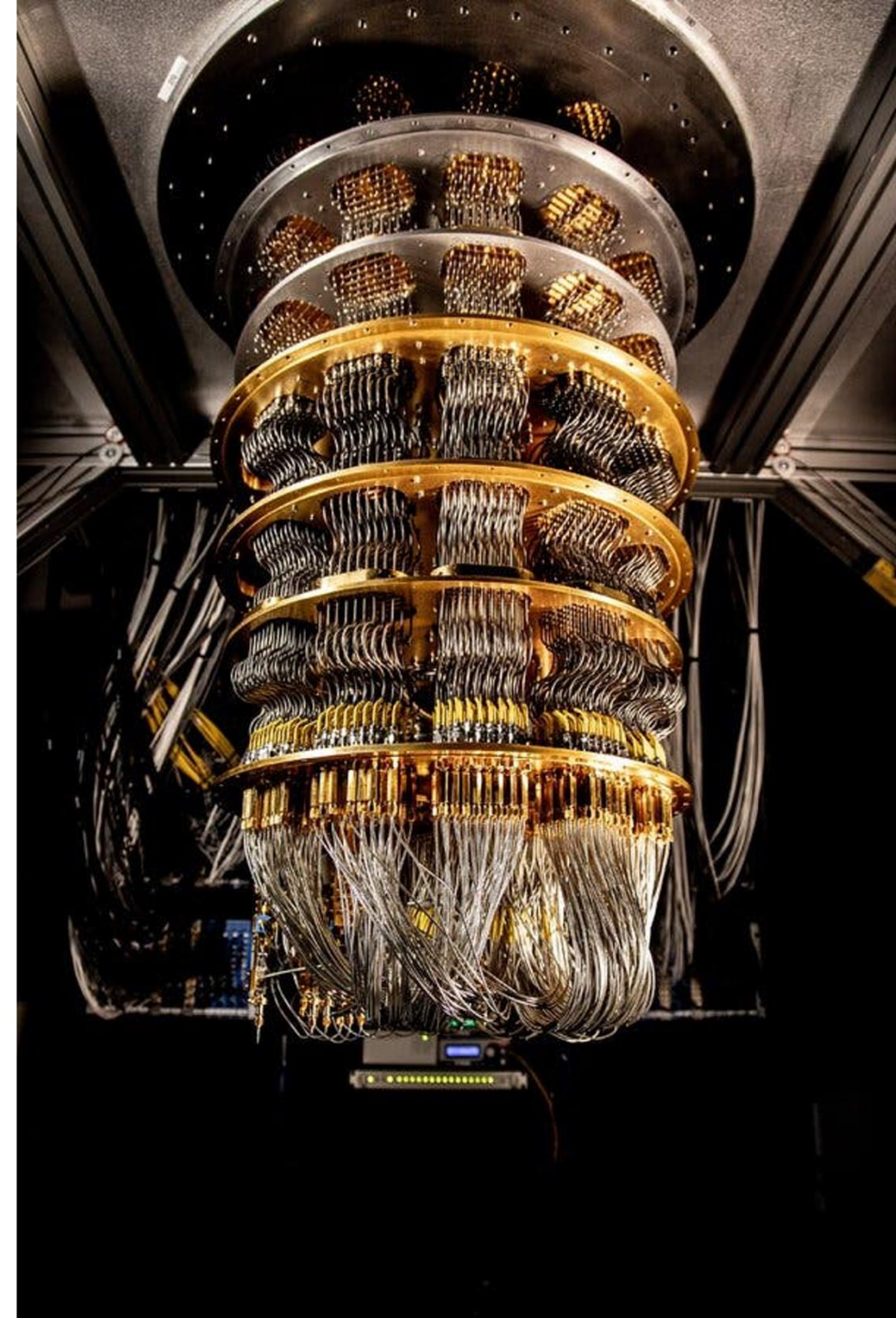
Inferring hidden quantum error states from noisy continuous measurements.



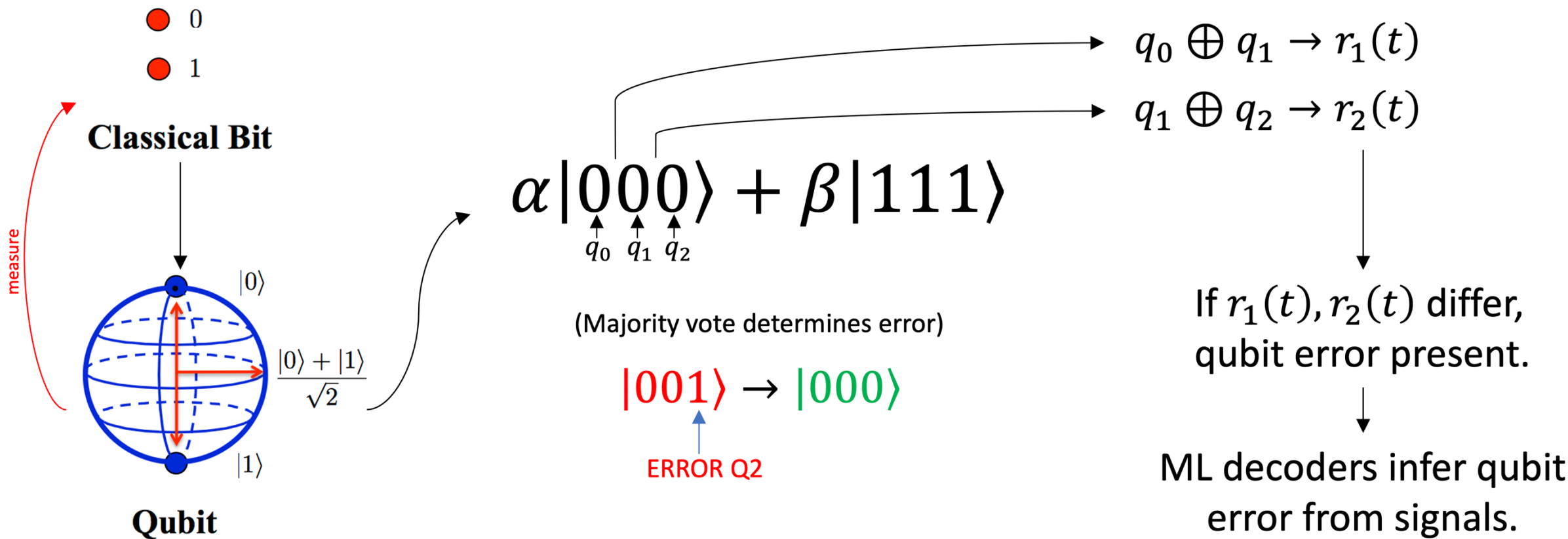
The Big Question:

Can **neural networks** outperform **Bayesian** decoders when real quantum hardware becomes messy and non-ideal?

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From Quantum States to Analog Signals



Quantum errors are hidden, so the decoder must infer them from continuous measurement signals.

Continuous analog measurements hide discrete quantum error states behind heavy noise.

Goal:

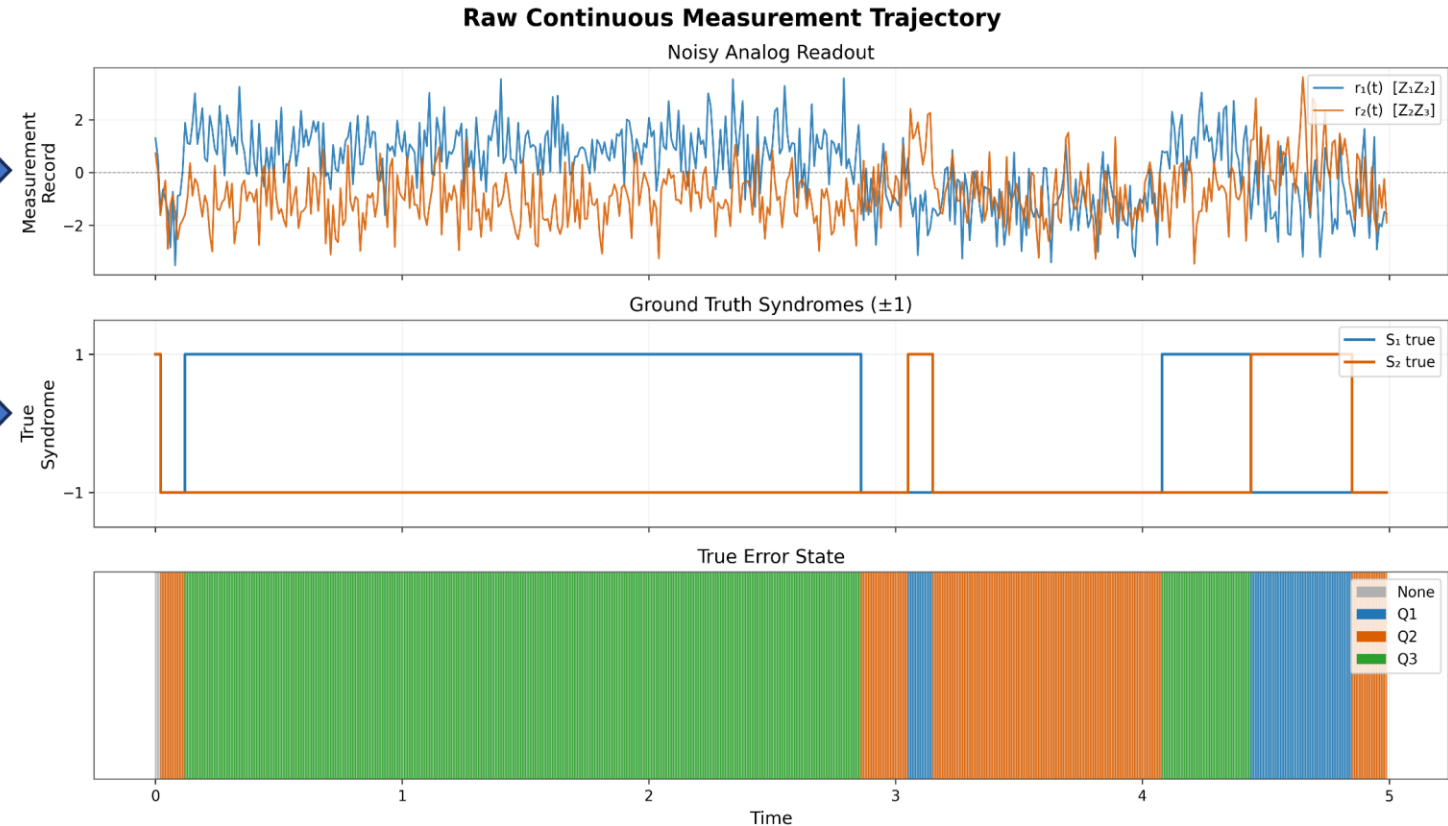
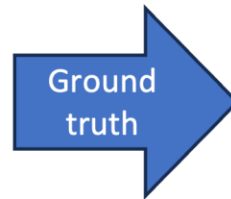
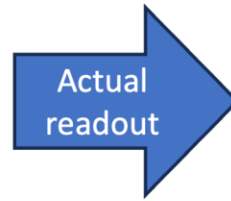
Monitor a 3-qubit repetition code continuously without collapsing the quantum state.

Challenge:

Signals are noisy analog trajectories, not clean digital bits.

Task:

Infer hidden error states from noisy streaming measurements.

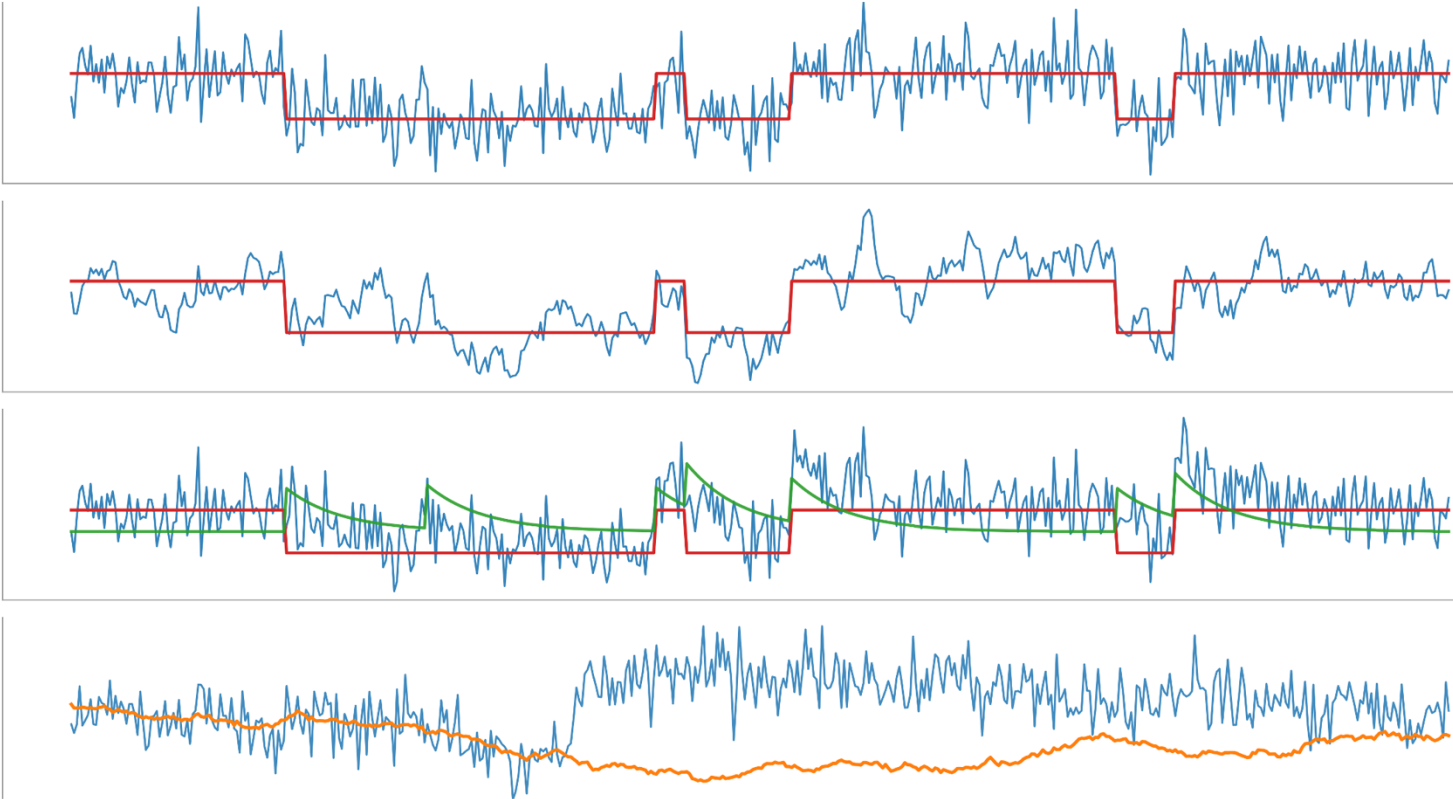


Measurement trajectories evolve sequentially. The data is a temporal stream rather than an isolated snapshot.

Increasing Hardware Realism

Each phase adds new non-ideal effects to the syndrome measurement process.

Phase	Added Effect
Phase 1	Gaussian noise baseline
Phase 2	Time-correlated noise
Phase 3	Transient ringing artifacts
Phase 4	Non-stationary drift



Each phase makes the readout less idealized and more hardware-like.

Bayesian Filter

A probabilistic state estimator that updates beliefs over time.

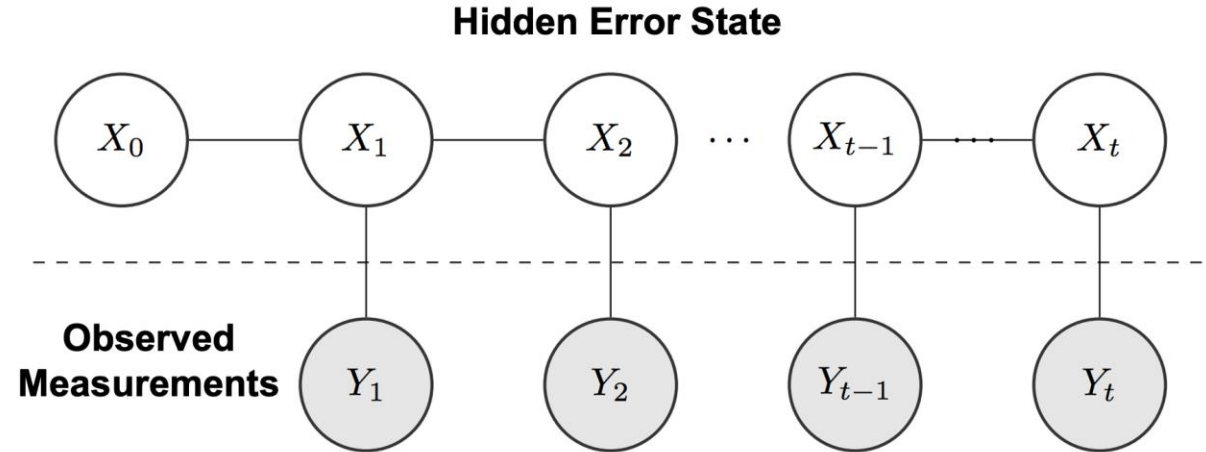
Core Idea

The Bayesian filter treats the true quantum error states as hidden variables. At each time step, the filter combines the previous belief state with the newest noisy measurement to recursively update a probability distribution over all possible error states as measurements arrive in real time.

Bayesian Update Equation

$$P(x_t | y_{1:t}) \propto P(y_t | x_t) P(x_t | y_{1:t-1})$$

Posterior belief $\propto$ measurement likelihood $\times$ prior belief



- Top row: hidden error states evolving over time.
- Bottom row: noisy measurements generated from those states.

Each new measurement incrementally refines the decoder's belief over the hidden error states.

Limitations of Bayesian Filtering

Bayesian filters perform well under ideal assumptions, but real quantum hardware rarely behaves exactly like the theoretical model.



Idealized Assumptions

- **White Gaussian Noise**
 - Independent noise
- **Fixed System Parameters**
 - Stationary calibration
- **Accurate Measurement Model**
 - Model matches hardware



Real Hardware Behavior

- **Colored Noise**
 - Time-correlated measurements
- **Drift**
 - Slowly shifting calibration
- **Transients**
 - Ringing after qubit flips
- **Measurement Backaction**
 - Measurements alter dynamics



As increasingly realistic hardware noise is introduced, Bayesian assumptions are violated. These effects accumulate over time and reduce Bayesian decoding performance, motivating the need for more adaptive sequence models like GRUs.

GRU Neural Network: Learned Sequence Modeling

A type of Recurrent Neural Network (RNN) specifically designed to handle sequences of data.

Core Idea

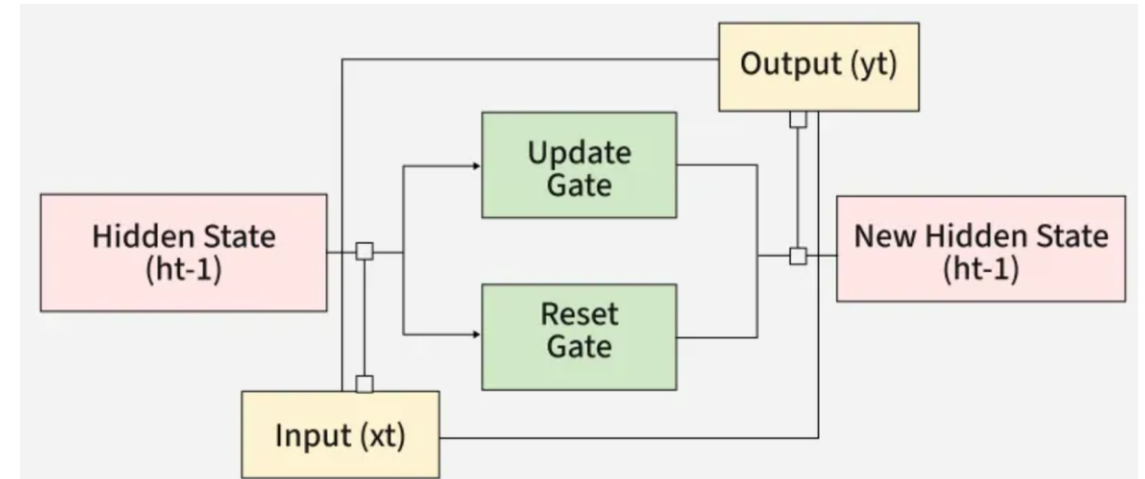
Basic GRUs use internal RESET and UPDATE gates to decide what information to keep from the past and what to discard as it receives new data.

Pseudo-Labeling

Pseudo-labeling is a “self-training” technique where the model treats its most confident predictions as the ground truth and learns from them.

Hybrid Labeling

“Goldilocks” solution. It combines the autonomy of pseudo-labeling with a tiny bit of “ground truth” to stop the model from a feedback loop of false positives.

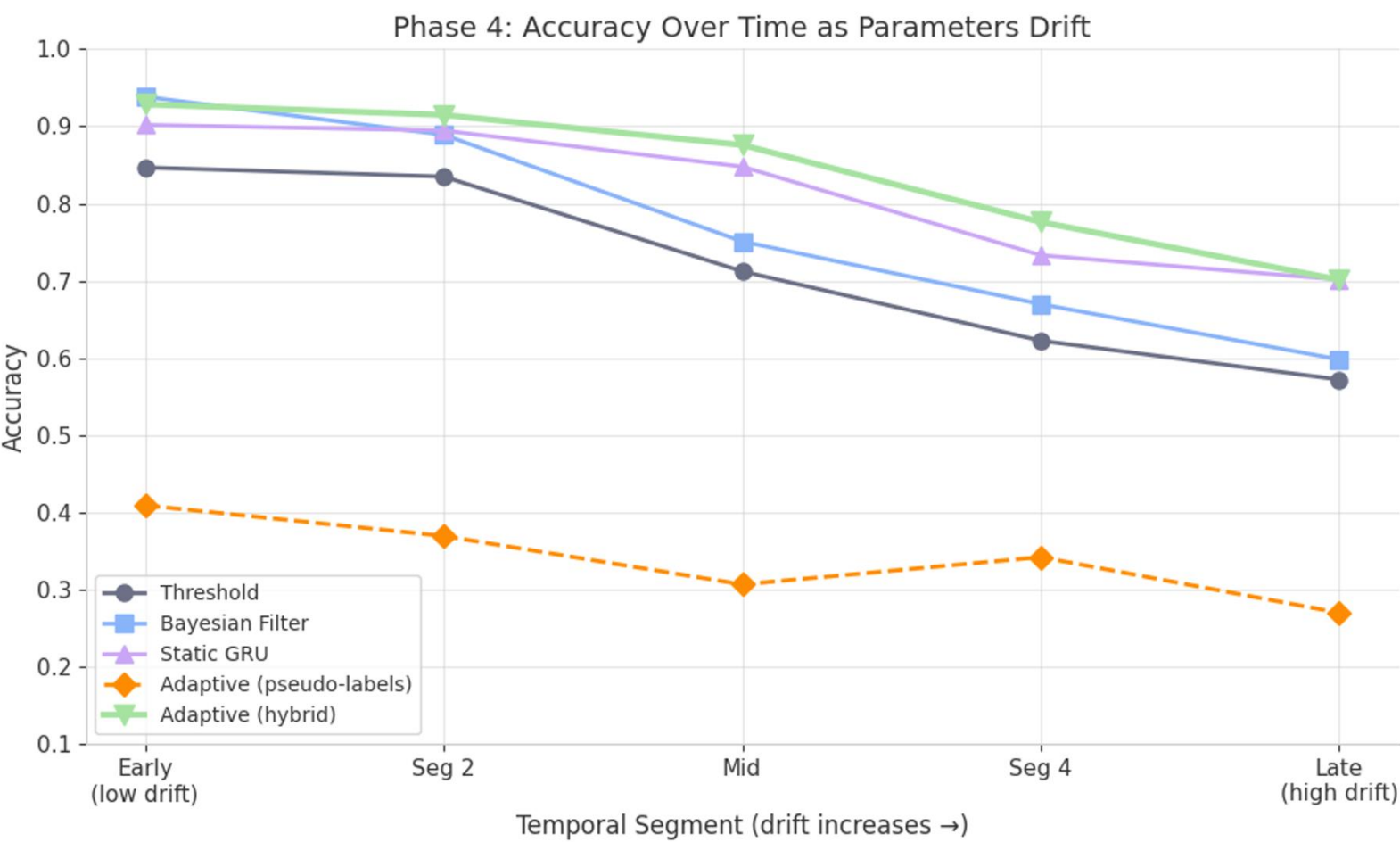


Sequential Memory Mechanism

- Hidden state stores temporal information over time
- Reset gate controls forgetting
- Update gate controls memory retention

Decoder Performance as Drift Increases

Bayesian filters perform best early, but adaptive GRUs remain stable as drift accumulates.



Early Drift

- **Bayesian filtering** performs best under **stationary noise**.
- **All decoders** begin with **high accuracy** under **near-ideal conditions**.

Increasing Drift

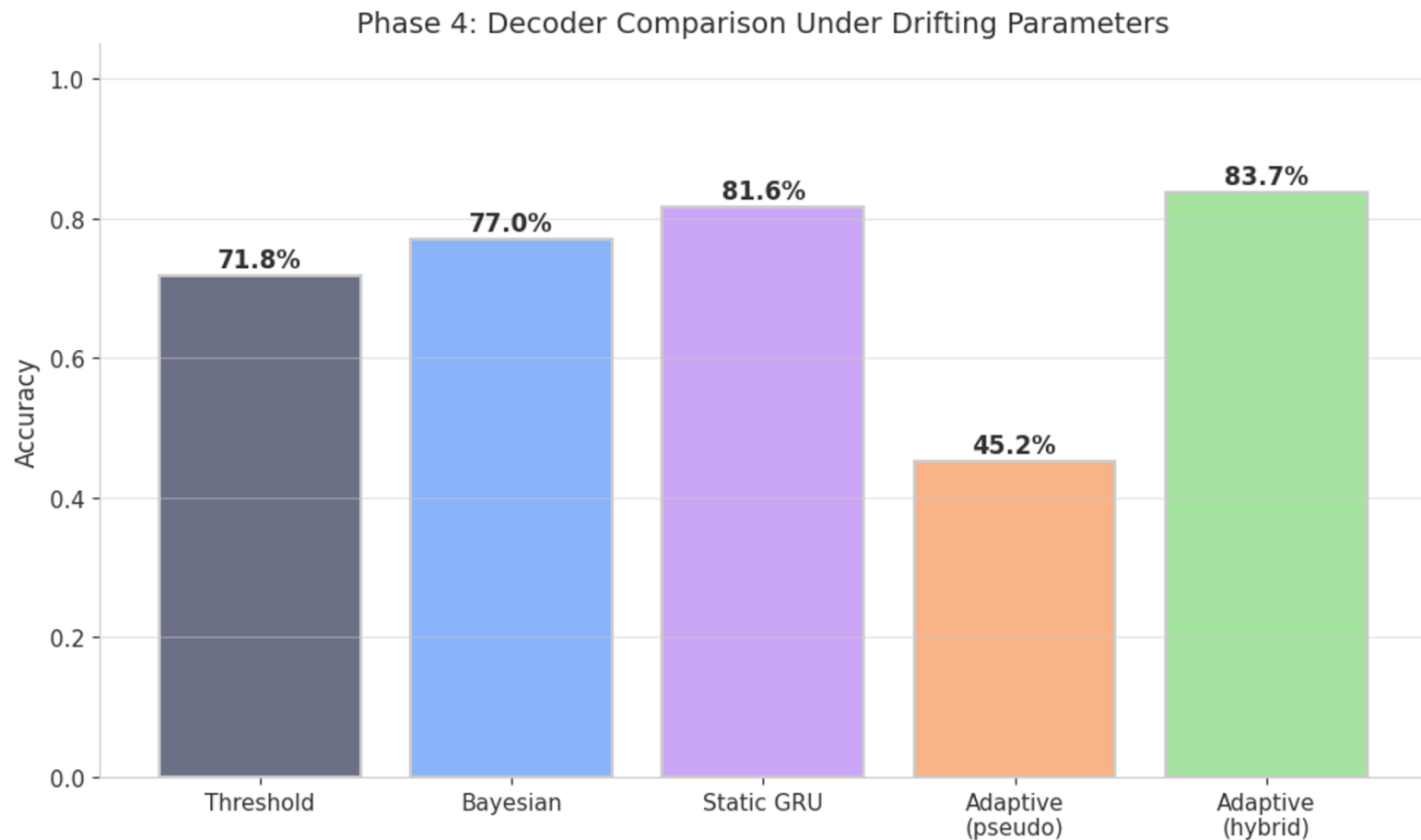
- **Bayesian** accuracy **declines** as **assumptions break**.
- **GRUs** better **preserve temporal structure** under drift.

Adaptive Learning

- Pure **pseudo-label collapses** from error accumulation.
- **Hybrid recalibration** remains **most stable** under sustained drift.

Adaptive Decoding Under Drift: Final Model Comparison

Hybrid recalibration GRU achieves the strongest overall robustness under drift.



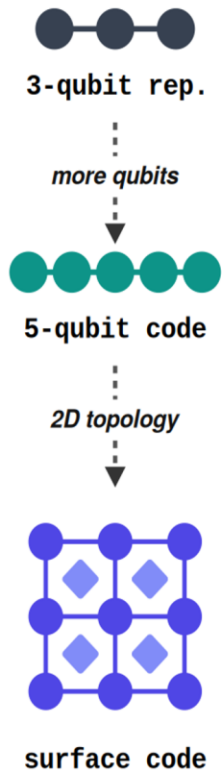
Overall Performance

- **Hybrid recalibration** achieves the **strongest overall robustness** under sustained drift.
- **Static GRUs** remain **more stable** than **Bayesian filtering** under realistic hardware conditions.
- **Threshold decoding** performs **worst** among the fixed decoders.
- **Pure pseudo-label** adaptation **collapses** from accumulated prediction errors.

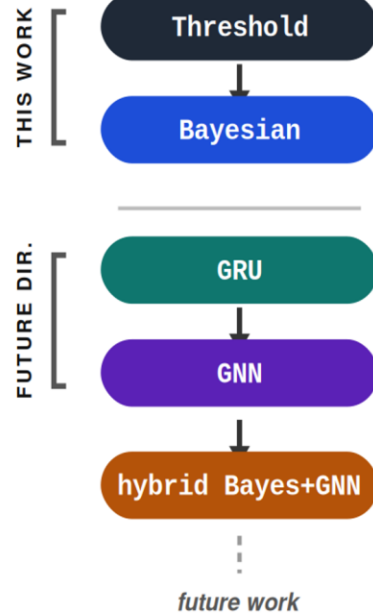
Learned sequence models outperform assumption-driven statistical decoders once realistic hardware non-idealities violate idealized assumptions.

The Road Ahead

CODE COMPLEXITY



DECODER



01

Temporal Structure

- Bayesian filters rely on fixed assumptions.
- GRUs learn temporal correlations directly from data.

03

Future Directions

- Graph neural networks for larger surface codes.
- Hybrid Bayes-neural decoders balancing interpretability and adaptability.
- Transfer learning from simulation to hardware.

02

Robustness Under Drift

- Drift and non-idealities violate Bayesian model assumptions.
- GRUs remain stable as hardware conditions evolve.



Final Takeaway: The best decoder is not the best equation — it is the model that best adapts to realistic quantum hardware.

Thank You

Research References

Convy, I., Liao, H., Zhang, S., et al.

Continuous quantum error correction for evolution under time-dependent Hamiltonians

Physical Review A, 103, 042406.

Convy, I., et al.

Machine Learning for Continuous Quantum Error Correction on Superconducting Qubits.

arXiv / Berkeley Quantum Computation Center.

Software & Tools

- Python
- PyTorch
- NumPy
- Matplotlib
- Jupyter Notebook
- Git & GitHub
- VSCode

Repository

github.com/pkarakala/cqec-ml-decoder